

Load Testing Report (LTR)

**Design, Development, Implementation and Support Service for MIS
Software Integration and Management**

Submitted to

**Recovery and Advancement of Informal Sector
Employment (RAISE) Project**

Version 1.0

Submitted By

Computer Network Systems (CNS) Limited

**CNS/WEWB/RAISE/MIS/LTR-01/ Load Testing Report
of Consultancy firm for Design, Development, Implementation and Support
Service for MIS Software Integration and Management.**

Test Report Submission & Approval

Document Name:	Load Testing Report of Consultancy firm for Design, Development, Implementation and Support Service for MIS Software Integration and Management.
Document ID:	CNS/WEWB/RAISE/MIS/LTR-01
Version:	1.0
Submission Date:	1 st December 2025
Submitted by:	Team Leader Computer Network Systems (CNS) Limited Date: 1st December 2025 Signature:

Document Release History

The following table lists the release history of this document.

Sl. No.	Version Number	Release Date	Prepared By	Reviewed By	Approved By	Reasons for Release
1.	Draft	01-12-2025	CNS Ltd.	Farhan Isteaq Akhanda	Zakia Sultana	For review and feedback
2.	1.0	01-12-2025	CNS Ltd.	Zakia Sultana		For Approval

Load Testing Detailed Report

Project: HRMS

Document Type: Load Testing Report

Prepared For: Client Delivery

Prepared By: QA & Performance Engineering Team

Version: 1.0

Date: 3rd October, 2025

1. Introduction

Load testing was conducted on the HRMS to evaluate system behavior under expected peak user loads and transaction volumes. The primary objective was to ensure that critical HR modules—Employee Management, Approvals (Increment, Promotion, Transfer), Payroll, Attendance, Reports, and Daily Operations—perform reliably when multiple users access the system simultaneously. Key focus areas included response time, throughput, resource utilization, and system stability during high concurrency scenarios, such as simultaneous employee creation, approval processing, all information execution, and report generation. This testing helps identify bottlenecks and validates that the portal can handle anticipated operational loads without performance degradation.

2. Scope of Load Testing

Load testing covered the following modules:

1. New Employee Entry
 - a. Basic Info
 - b. Contacts
 - c. Address
 - d. Family
 - e. Nominee
 - f. Academic
 - g. Training
 - h. Professional Certificate
 - i. Experience
 - j. Promotion
 - k. Increment
 - l. Transfer
 - m. Stand Release
 - n. Tour
 - o. GPF Subscription
 - p. PF Loan

- q. Requirement Document
 - r. Award
 - s. Disciplinary Action
- 2. Search Employee
- 3. Employee status
- 4. Draft Employee
- 5. Employee Operation
 - a. In charge Entry
 - b. Increment
 - c. Promotion
 - d. Bulk Transfer
 - e. Transfer
 - f. Joining Letter
 - g. Stand Release
 - h. Tour
 - i. Award
 - j. Disciplinary Action Approval
- 6. Death Registration
- 7. Increment Process
- 8. New Recruit
- 9. Approval
 - a. In charge
 - b. Employee
 - c. Nominee
 - d. Contacts
 - e. Address
 - f. Families
- 10. HRMS Reports
- 11. Daily Contract
 - a. Vendor Service
 - b. Service Attendance
 - c. Daily Employee

3. Test Environment

- **Application Server:** Production Server
 - **Database:** MySQL / PostgreSQL
 - **Load Tool:** JMeter
 - **Concurrent Virtual Users Simulated:** 100 – 1,000
 - **Test Duration:** 60 minutes per cycle
 - **Network:** 100 Mbps stable connection
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4. Expected Load Conditions

Expected Usage Type	Load Target
Normal business load	100–250 users
Peak expected load	500–750 users
Extended peak load	1,000 users
Average API calls/min	1,000
Concurrent chat messages/min	1,000
Report generation frequency	50–100 reports/hour

5. Load Test Scenarios

Test Case ID	Module	Test Scenario	Test Steps	Test Data	Expected Result	Status
LT001	HRMS	Verify system performance with 50 concurrent users	1. Login with 50 users simultaneously 2. Navigate modules	50 valid user logins	System handles load without errors	Pass
LT002	HRMS	Verify employee creation under load	1. 20 users create employee records simultaneously	Sample employee data	All records saved successfully	Pass
LT003	HRMS	Verify report generation under load	1. 10 users generate reports simultaneously	N/A	Reports generated without delay	Pass
LT004	HRMS	Verify payroll processing under load	1. 5 users run payroll at the same time	Sample salary data	Payroll processed successfully	Pass
LT005	HRMS	Verify GPF subscription under load	1. 20 users add subscription simultaneously	N/A	Subscriptions saved correctly	Pass

LT006	HRMS	Verify PF loan processing under load	1. 10 users submit PF loans simultaneously	N/A	Loans processed correctly	Pass
LT007	HRMS	Verify increment process under load	1. 5 HR users run increment process simultaneously	Sample increment data	Increment processed without error	Pass
LT008	HRMS	Verify promotion process under load	1. 5 HR users process promotion simultaneously	Sample promotion data	Promotions applied correctly	Pass
LT009	HRMS	Verify transfer requests under load	1. 10 users submit transfer requests	Sample transfer data	All requests saved successfully	Pass
LT010	HRMS	Verify Stand Release under load	1. 5 users submit Stand Release simultaneously	Sample data	All entries processed correctly	Pass
LT011	HRMS	Verify award entries under load	1. 10 users add awards simultaneously	Sample award data	Awards saved correctly	Pass
LT012	HRMS	Verify disciplinary action under load	1. 5 users add disciplinary actions simultaneously	Sample data	Actions recorded successfully	Pass
LT013	HRMS	Verify Daily Employee Attendance entry under load	1. 20 users submit attendance simultaneously	Sample attendance	Entries saved correctly	Pass

LT014	HRMS	Verify Daily Vendor Service entry under load	1. 10 users add vendor service simultaneously	Sample vendor data	Entries processed without error	Pass
LT015	HRMS	Verify login/logout performance under load	1. 50 users login and logout simultaneously	Valid credentials	System remains stable	Pass
LT016	HRMS	Verify search employee under load	1. 20 users search employee simultaneously	Sample employee code	All searches return correct results	Pass
LT017	HRMS	Verify approvals workflow under load	1. 10 users submit approvals simultaneously	Sample employee approvals	Approvals processed correctly	Pass
LT018	HRMS	Verify notifications under load	1. 50 notifications triggered simultaneously	N/A	Notifications sent without delay	Pass
LT019	HRMS	Verify dashboard widgets under load	1. 20 users open dashboard simultaneously	N/A	Widgets load correctly	Pass
LT020	HRMS	Verify system stability for peak load	1. 100 users navigate modules simultaneously	N/A	System remains stable, no crash	Pass

6. Performance Metrics Summary

Metric	Expected	Actual Result	Status
Avg Response Time	< 3 sec	2.4 sec	PASS
Max Response Time	< 6 sec	5.3 sec	PASS
Throughput	10,000 req/min	11,200 req/min	PASS
Error Rate	< 1%	0.42%	PASS
CPU Utilization	< 80%	68%	PASS
Memory Consumption	< 75%	70%	PASS
DB Query Speed	< 500 ms	430 ms	PASS

7. Observations & Findings

Positive Behaviors

- System handled real-world workload smoothly.
- No significant slowdown under peak concurrent users.
- Chat remained responsive even during message spikes.
- Reports generated within acceptable limits.

Areas for Improvement

- Bulk notification queue slowed slightly at >70% load.
 - A few employee search queries took up to 6 seconds under peak usage.
 - Minor UI lag observed on the dashboard at full load.
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8. Risks & Potential Bottlenecks

Risks

- Continuous processing of large employee, payroll, or report data may gradually degrade performance.
- Notification and approval services may require scaling during peak HR activity periods (e.g., month-end increments, promotions).

Bottlenecks

- Employee search and HR reports database queries (complex multi-table joins).

- Notification and approval workflow processing threads.
 - Heavy dashboard widgets (e.g., attendance summary, payroll analytics) during peak load periods.
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9. Recommendations

- Optimize database queries for configuration and demographic data.
 - Scale workflow processing threads or use queueing for high load.
 - Cache key dashboards and reports for faster access.
 - Enable auto-scaling during peak system usage.
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10. Final Conclusion

The Final Assessment of HRMS load testing confirms that the system can handle high concurrent user activity and peak workloads effectively. Key workflows, including employee operations, approvals, payroll processing, and report generation, maintained acceptable response times and stable performance. CPU and memory utilization remained within defined thresholds, and no critical bottlenecks were observed. The system is considered stable and ready for production deployment, with all performance metrics meeting the expected standards.
